

Rigid Body Rotations - Approved Segments

53 approved segments, in reading order

SUMMARY ON ELEMENTARY ROTATIONS:

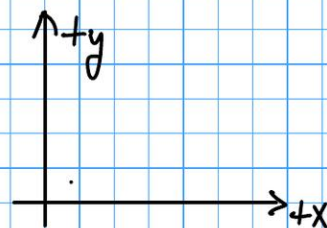
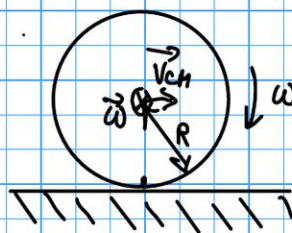
$$L_z = I \omega_z \quad (1) \quad \vec{N} = \vec{r} \times \vec{F} \quad (2) \quad T = \frac{I}{2} \omega_z^2 = \frac{\omega_z L_z}{2} \quad (3)$$

$$\frac{d\vec{L}}{dt} = \vec{N}(\vec{e}) \quad (4) \quad \left\{ \begin{array}{l} \text{INERTIAL FRAME} \\ \text{CM FRAME (ORIGIN AT CM,} \\ \text{AXES PARALLEL TO INERTIAL AXES)} \end{array} \right.$$

Rolling without
slipping condition:

$$v_{Cx} + \omega_z R = 0 \quad (5)$$

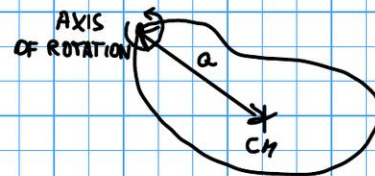
($\omega_z < 0$ for clockwise rotation)
(6)



Parallel Axis Theorem: $I = \bar{I} + Ma^2$ (7)

(8) $\bar{I} \equiv$ Moment of inertia around CM

(9) $a \equiv$ Distance between axis of rotation and CM



(9) a = distance between axis of rotation and CM

Separating CM Motion:

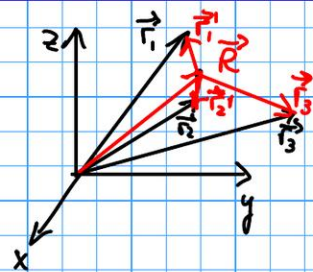
$$M = \sum m_j \quad (12)$$

$$\vec{L} = \vec{R} \times M \dot{\vec{R}} + \sum_j \vec{r}_j' \times m_j \dot{\vec{r}}_j' \quad (10)$$

$$\vec{R} = \frac{\sum m_j \vec{r}_j}{M} \quad (13)$$

$$T = \frac{M}{2} |\dot{\vec{R}}|^2 + \frac{1}{2} \sum_j m_j |\dot{\vec{r}}_j'|^2 \quad (11)$$

$$\vec{r}_j' = \vec{r}_j - \vec{R} \quad (14) \quad \text{POSITION RELATIVE TO CM}$$



For rigid body, considering rotation with angular velocity ω_z around CM:

$$L_z = (\vec{R} \times M \dot{\vec{R}})_z + I \omega_z \quad (16)$$

$$T = \frac{M}{2} |\dot{\vec{R}}|^2 + \frac{I}{2} \omega_z^2 \quad (17)$$

PROOFS:

$$\Rightarrow - \sum_{i=1}^N \vec{r}_i \cdot \nabla m_i \cdot \dot{\vec{r}}_i = \sum_{i=1}^N (\vec{R} + \vec{r}_i') \times m_i (\dot{\vec{R}} + \dot{\vec{r}}_i') = \sum \vec{R} \times m_i \dot{\vec{R}} + \sum \vec{r}_i' \times m_i \dot{\vec{r}}_i' + \text{CROSS-TERMS} \quad (18)$$

$$L_2 = \frac{1}{2} \sum_{i=1}^N m_i \dot{\vec{r}}_i^2 = \frac{1}{2} M \dot{\vec{R}}^2 + \frac{1}{2} \sum_{i=1}^N m_i \dot{\vec{r}}_i'^2$$

PROOFS:

CHANGING TO
CM SYSTEM:

$$L = \sum_{i=1}^N \vec{r}_i \times m_i \dot{\vec{r}}_i = \sum_{i=1}^N (\vec{R} + \vec{r}_i') \times m_i (\dot{\vec{R}} + \dot{\vec{r}}_i') = \sum_i \vec{R} \times m_i \dot{\vec{R}} + \sum_i \vec{r}_i' \times m_i \dot{\vec{r}}_i' + \text{CROSS-TERMS} \quad (18)$$

$$\vec{r}_i = \vec{R} + \vec{r}_i' \quad \dot{\vec{r}}_i = \dot{\vec{R}} + \dot{\vec{r}}_i' \quad (19)$$

$$\text{CROSS-TERMS} \left\{ \begin{aligned} \sum_i \vec{r}_i' \times m_i \dot{\vec{R}} &= \sum_i m_i (\vec{r}_i' - \vec{R}) \times \dot{\vec{R}} = \left[\sum_i m_i \vec{r}_i' - \left(\sum_i m_i \right) \vec{R} \right] \times \dot{\vec{R}} = [M\dot{\vec{R}} - M\dot{\vec{R}}] \times \dot{\vec{R}} = \vec{0} \quad (20) \\ \sum_i \vec{R} \times m_i \dot{\vec{r}}_i' &= \vec{R} \times \sum_i m_i (\dot{\vec{r}}_i' - \dot{\vec{R}}) = \vec{R} \times \left[\sum_i m_i \dot{\vec{r}}_i' - \left(\sum_i m_i \right) \dot{\vec{R}} \right] = \vec{R} \times [M\dot{\vec{R}} - M\dot{\vec{R}}] = \vec{0} \quad (21) \end{aligned} \right.$$

THEN:

$$L = \vec{R} \times M \dot{\vec{R}} + \sum_i \vec{r}_i' \times m_i \dot{\vec{r}}_i' \quad (22)$$

$$\Rightarrow \frac{dL}{dt} = \vec{R} \times M \ddot{\vec{R}} + \sum_i \vec{r}_i' \times m_i \ddot{\vec{r}}_i' \quad (23) \quad \text{TIME DERIVATIVE}$$

THEN: $\vec{L} = \vec{R} \times M \dot{\vec{R}} + \sum_i \vec{r}_i' \times m_i \dot{\vec{r}}_i' \quad (22)$

BUT ALSO: $\frac{d\vec{L}}{dt} = \vec{N}^{(e)} = \sum_i \vec{r}_i \times \vec{F}_i^{(e)} \quad (24)$

$\Rightarrow \frac{d\vec{L}}{dt} = \vec{R} \times M \ddot{\vec{R}} + \underbrace{\sum_i \vec{r}_i' \times m_i \ddot{\vec{r}}_i'}_{d\vec{L}'/dt} \quad (23) \quad \text{TIME DERIVATIVE}$

$\frac{d\vec{L}}{dt} = \vec{R} \times \sum_i \vec{F}_i^{(e)} + \sum_i \vec{r}_i' \times \vec{F}_i^{(e)} = \vec{R} \times M \ddot{\vec{R}} + \sum_{i=1}^N \vec{r}_i' \times \vec{F}_i^{(e)} \quad (26)$

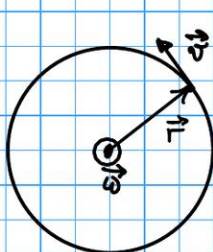
$(25) M \ddot{\vec{R}} = \sum_i \vec{F}_i^{(e)}$ Newton's 2nd law for particle system

$\frac{d\vec{L}'}{dt} = \sum_{i=1}^N \vec{r}_i' \times \vec{F}_i^{(e)} \quad (27)$

Eq (27) is the special case of Eq (4) in the CM Frame.

Rotations, Non-inertial systems, and Rigid body motion

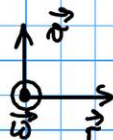
CIRCULAR MOTION
IN A PLANE



$$\vec{v} = \vec{\omega} \times \vec{r}$$

$$\vec{v} \perp \vec{r}$$

$$\vec{v} \cdot \vec{r} = 0$$



$$\hat{e}_i \cdot \hat{e}_j = \delta_{ij} = \begin{cases} 1 & i=j \\ 0 & i \neq j \end{cases}$$

ROTATING COORDINATE SYSTEM



$$0 = \frac{d\delta_{ij}}{dt} = \frac{d\hat{e}_i \cdot \hat{e}_j}{dt}$$

$$0 = \hat{e}_i \cdot \frac{d\hat{e}_j}{dt} \quad i=j$$

$$0 = \hat{e}_i \cdot \frac{d\hat{e}_j}{dt} + \hat{e}_j \cdot \frac{d\hat{e}_i}{dt} \quad i \neq j$$

$(\hat{e}_j)_{j=1,2,3}$ IS A BASIS $\Rightarrow$ WE CAN WRITE $\frac{d\hat{e}_i}{dt}$ AS A LINEAR COMBINATION

$$\frac{d\hat{e}_i}{dt} = \sum_{j=1}^3 \Omega_{ij} \hat{e}_j \quad \Omega_{ij} = \hat{e}_j \cdot \frac{d\hat{e}_i}{dt}$$

(USE THAT $\hat{e}_j \cdot \sum_{i=1}^3 \Omega_{ij} \hat{e}_i = \sum_{i=1}^3 \Omega_{ij} \hat{e}_j \cdot \hat{e}_i = \Omega_{ij}$)

$$\hat{x} \times \hat{y} = \hat{z}$$

$$\hat{y} \times \hat{z} = \hat{x}$$

$$\hat{z} \times \hat{x} = \hat{y}$$

$$\hat{e}_1 \equiv \hat{x} \quad \hat{e}_2 \equiv \hat{y} \quad \hat{e}_3 \equiv \hat{z}$$

$$\hat{e}_1 \times \hat{e}_2 = \hat{e}_3 \quad (\text{AND CYCLICALLY})$$

$$\hat{e}_2 \times \hat{e}_3 = \hat{e}_1$$

$$\hat{e}_3 \times \hat{e}_1 = \hat{e}_2$$

Levi-Civita symbol

$$\epsilon_{ijk} = \begin{cases} +1 & ijk \text{ cyc } 123 \\ -1 & ijk \text{ cyc } 321 \\ 0 & \text{if any repeated index} \end{cases}$$

$$\epsilon_{123} = \epsilon_{231} = \epsilon_{312} = +1$$

$$\epsilon_{213} = \epsilon_{132} = \epsilon_{321} = -1$$

$$\epsilon_{jji} = \epsilon_{ijj} = \epsilon_{iji} = 0$$

$$\hat{e}_i \times \hat{e}_j = \sum_k \epsilon_{ijk} \hat{e}_k$$

$$\langle \varphi_i | \left(\sum_j c_j | \varphi_j \rangle \right) = \sum_j c_j \langle \varphi_i | \varphi_j \rangle = c_i$$

$i \neq j$

$$0 = \Omega_{ij} + \Omega_{ji} \Rightarrow \Omega_{ji} = -\Omega_{ij}$$

$$\Omega_{ii} = \hat{e}_i \cdot \frac{d\hat{e}_i}{dt} = 0$$

$$\begin{pmatrix} 0 & \Omega_{12} & -\Omega_{31} \\ -\Omega_{12} & 0 & \Omega_{23} \\ \Omega_{31} & -\Omega_{23} & 0 \end{pmatrix}$$

$$\omega_1 = \Omega_{23} \quad \omega_2 = \Omega_{31} \quad \omega_3 = \Omega_{12}$$

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at

$$\frac{d\hat{e}_1}{dt} = \Omega_{12} \hat{e}_2 + \Omega_{13} \hat{e}_3 = \Omega_{12} \hat{e}_2 - \Omega_{31} \hat{e}_3 = \omega_3 \hat{e}_2 - \omega_2 \hat{e}_3 = \vec{\omega} \times \hat{e}_1 \quad \checkmark$$

$$\vec{\omega} \times \hat{e}_1 = (\omega_1 \hat{e}_1 + \omega_2 \hat{e}_2 + \omega_3 \hat{e}_3) \times \hat{e}_1 = -\omega_2 \hat{e}_3 + \omega_3 \hat{e}_2$$

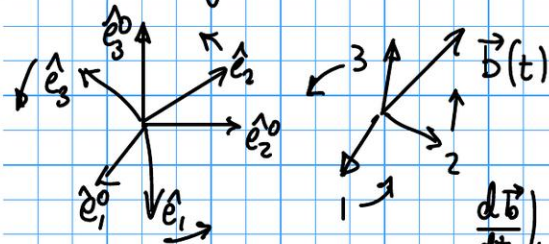
$$\frac{d\hat{e}_2}{dt} = \Omega_{21} \hat{e}_1 + \Omega_{23} \hat{e}_3 = -\Omega_{12} \hat{e}_1 + \Omega_{31} \hat{e}_3 = -\omega_3 \hat{e}_1 + \omega_1 \hat{e}_3 = \vec{\omega} \times \hat{e}_2 \quad \checkmark$$

$$\vec{\omega} \times \hat{e}_2 = (\omega_1 \hat{e}_1 + \omega_2 \hat{e}_2 + \omega_3 \hat{e}_3) \times \hat{e}_2 = \omega_1 \hat{e}_3 - \omega_3 \hat{e}_1$$

$$\hat{e}_1 \times \hat{e}_2 = \hat{e}_3 \quad \hat{e}_3 \times \hat{e}_2 = -\hat{e}_1$$

$$\frac{d\hat{e}_3}{dt} = \vec{\omega} \times \hat{e}_3$$

Rotating reference frames



Fixed inertial reference frame $\hat{e}_1^0, \hat{e}_2^0, \hat{e}_3^0$

Rotating (non-inertial) reference frame $\hat{e}_1^i, \hat{e}_2^i, \hat{e}_3^i$

$$\left. \frac{d\vec{b}}{dt} \right|_{\text{inertial}} = \frac{d}{dt} \left(\sum_{i=1}^3 b_i^0 \hat{e}_i^0 \right) = \sum_{i=1}^3 \dot{b}_i^0 \hat{e}_i^0$$

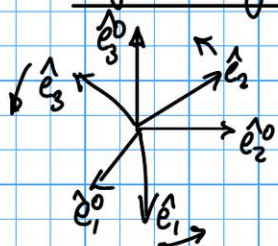
$$\left. \frac{d\vec{b}}{dt} \right|_{\text{inertial}} = \frac{d}{dt} \left(\sum_{i=1}^3 b_i \hat{e}_i \right) = \sum_{i=1}^3 \dot{b}_i \hat{e}_i + \sum_{i=1}^3 b_i \dot{\hat{e}}_i \quad \left. \frac{d\vec{b}}{dt} \right|_{\text{body}} \equiv \sum_{i=1}^3 \frac{db_i}{dt} \hat{e}_i$$

$$\dot{\hat{e}}_i = \frac{d\hat{e}_i}{dt} = \vec{\omega} \times \hat{e}_i$$

For rotations $\left. \frac{d\vec{b}}{dt} \right|_{\text{inertial}} = \left. \frac{d\vec{b}}{dt} \right|_{\text{body}} + \sum_{i=1}^3 b_i (\vec{\omega} \times \hat{e}_i)$

$$\boxed{\left. \frac{d\vec{b}}{dt} \right|_{\text{inertial}} = \left. \frac{d\vec{b}}{dt} \right|_{\text{body}} + \vec{\omega} \times \vec{b}}$$

Rigid body rotations



$\vec{r}_p$: position of point p in the body frame (CONSTANT IN BODY FRAME)
Rigid body with one fixed point.

$$T = \sum_p \frac{1}{2} m_p v_p^2 = \sum_p \frac{1}{2} m_p (\vec{\omega} \times \vec{r}_p)^2$$

Levi-Civita symbol

$$\epsilon_{ijk} = \begin{cases} +1 & \text{ijk cyc 123} \\ -1 & \text{ijk cyc 213} \\ 0 & \text{if any repeated index} \end{cases}$$

$$\epsilon_{123} = \epsilon_{231} = \epsilon_{312} = +1$$

$$\epsilon_{213} = \epsilon_{132} = \epsilon_{321} = -1$$

$$\epsilon_{dii} = \epsilon_{ijj} = \epsilon_{jij} = 0$$

$$\hat{e}_1 \times \hat{e}_2 = \hat{e}_3 = \sum_l \epsilon_{12l} \hat{e}_l \checkmark$$

$$\sum_l \epsilon_{12l} \hat{e}_l = \hat{e}_3$$

$$\hat{e}_i \times \hat{e}_j = \sum_l \epsilon_{ijl} \hat{e}_l \left(\begin{smallmatrix} \text{notation} \end{smallmatrix} \right)$$

$$\vec{a} \times \vec{b} = \left(\sum_i a_i \hat{e}_i \right) \times \left(\sum_j b_j \hat{e}_j \right) = \sum_{ijl} a_i b_j \epsilon_{ijl} \hat{e}_l = \sum_{ijk} \epsilon_{ijk} a_i b_j \hat{e}_k \quad (\vec{a} \times \vec{b})_k = \sum_{ij} \epsilon_{ijk} a_i b_j$$

Also

$$\sum_i \epsilon_{jki} \epsilon_{lmk} = \sum_i \epsilon_{ijk} \epsilon_{ilm} = \delta_{jl} \delta_{km} - \delta_{jm} \delta_{kl}$$

$i \leftarrow j \quad k \quad l \quad m \quad n \quad (\text{cyclic})$

$$\begin{array}{c} i \\ \circlearrowleft \\ j \end{array} \quad \epsilon_{ijk} = \epsilon_{jki} = \epsilon_{kij} \quad (\text{CYCLIC})$$

$$\begin{aligned}
 (\vec{A} \times \vec{B}) \cdot (\vec{C} \times \vec{D}) &= \left(\sum_{ijk} A_i B_j \epsilon_{ijk} \hat{e}_k \right) \cdot \left(\sum_{lmn} C_l D_m \epsilon_{lmn} \hat{e}_n \right) = \sum_{ijklm} \epsilon_{ijk} \epsilon_{lmk} A_i B_j C_l D_m \\
 &= \sum_{ijklm} (\delta_{il} \delta_{jm} - \delta_{im} \delta_{jl}) A_i B_j C_l D_m = (\vec{A} \cdot \vec{C})(\vec{B} \cdot \vec{D}) - (\vec{A} \cdot \vec{D})(\vec{B} \cdot \vec{C})
 \end{aligned}$$

IN SUMMARY:

$$T = \sum_P \frac{1}{2} m_P v_P^2 = \sum_P \frac{1}{2} m_P (\vec{\omega} \times \vec{r}_P)^2 = \sum_P \frac{m_P}{2} [\omega^2 r_P^2 - (\vec{\omega} \cdot \vec{r}_P)^2]$$

$$(\vec{A} \times \vec{B}) \cdot (\vec{C} \times \vec{D}) = (\vec{A} \cdot \vec{C})(\vec{B} \cdot \vec{D}) - (\vec{A} \cdot \vec{D})(\vec{B} \cdot \vec{C}) \Rightarrow (\vec{\omega} \times \vec{r}_P) \cdot (\vec{\omega} \times \vec{r}_P) = (\vec{\omega} \cdot \vec{\omega})(\vec{r}_P \cdot \vec{r}_P) - (\vec{\omega} \cdot \vec{r}_P)(\vec{\omega} \cdot \vec{r}_P)$$

$$T = \frac{1}{2} \sum_p m_p \left[r_p^2 \sum_{i=1}^3 \omega_i \omega_i - \left(\sum_{i=1}^3 \omega_i r_{pi} \right) \left(\sum_{j=1}^3 \omega_j r_{pj} \right) \right] = \frac{1}{2} \sum_p m_p \sum_{ij=1}^3 \left[r_p^2 \delta_{ij} - r_{pi} r_{pj} \right] \omega_i \omega_j$$

$$T = \frac{1}{2} \sum_{ij=1}^3 I_{ij} \omega_i \omega_j$$

$$I_{ij} \equiv \sum_p m_p \left[r_p^2 \delta_{ij} - r_{pi} r_{pj} \right]$$

Moment of inertia
tensor (SYMMETRIC)

$$\vec{L} = \sum_p m_p \vec{r}_p \times \vec{v}_p = \sum_p m_p \vec{r}_p \times (\vec{\omega} \times \vec{r}_p)$$

Aside: Using summation notation:

$$\begin{aligned} [A \times (B \times C)]_i &= \epsilon_{ijk} A_j \epsilon_{k\ell m} B_\ell C_m = \epsilon_{kij} \epsilon_{k\ell m} A_j B_\ell C_m = (\delta_{i\ell} \delta_{jm} - \delta_{im} \delta_{j\ell}) A_j B_\ell C_m \\ &= (\vec{A} \cdot \vec{C}) B_i - (\vec{A} \cdot \vec{B}) C_i \end{aligned}$$

$$[\vec{r}_p \times (\vec{\omega} \times \vec{r}_p)]_i = (\vec{r}_p \cdot \vec{r}_p) \omega_i - (\vec{r}_p \cdot \vec{\omega}) r_{pi}$$

$$L_i = \sum_p m_p (\vec{r}_p \times \vec{v}_p)_i = \sum_p m_p [\vec{r}_p \times (\vec{\omega} \times \vec{r}_p)]_i = \sum_p m_p \left[r_p^2 \omega_i - \sum_{j=1}^3 r_{pi} r_{pj} \omega_j \right]$$

$$L_i = \sum_{j=1}^3 \left(\underbrace{\sum_p m_p \left[r_p^2 \delta_{ij} - r_{pi} r_{pj} \right]}_{= I_{ij}} \right) \omega_j = \sum_{j=1}^3 I_{ij} \omega_j$$

$$L_i = \sum_j I_{ij} \omega_j$$

$$T = \frac{1}{2} \vec{\omega} \cdot \vec{L}$$

$$T = \frac{1}{2} \sum_{i,j=1}^3 I_{ij} \omega_i \omega_j$$

$$I_{ij} = I_{ji} = I_{ji}^* \Rightarrow \text{HERMITIAN} \Rightarrow \text{CAN BE DIAGONALIZED} \quad \begin{pmatrix} I_1 & 0 & 0 \\ 0 & I_2 & 0 \\ 0 & 0 & I_3 \end{pmatrix}$$

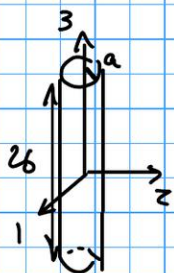
Directions that diagonalize I_{ij} are called PRINCIPAL AXES

EXAMPLES

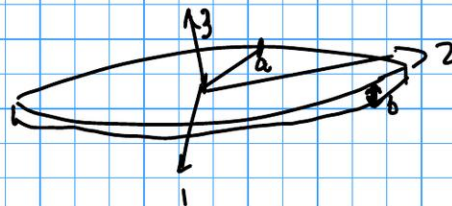
Homogeneous CUBE OR SPHERE



$$\begin{pmatrix} I & 0 & 0 \\ 0 & I & 0 \\ 0 & 0 & I \end{pmatrix} = I_{3 \times 3}$$



$$a \ll b \\ I_1 = I_2 > I_3$$



$$a \gg b \\ I_1 = I_2 < I_3$$

$$I = \sum_p m_p (\text{distance to axis})^2 \\ = \sum_p m_p r_{p1}^2$$

$$I_3 \sim \# M a^2 \quad I_1 \sim \# M b^2$$

$$a \ll b \Rightarrow I_1 \ll I_3$$

$$a \gg b \Rightarrow I_3 \gg I_1$$

$$I_{ij} = \sum_p m_p [\delta_{ij} r_p^2 - r_{pi} r_{pj}]$$

$$I_{33} = \sum_p m_p [\delta_{33} r_p^2 - r_{p3} r_{p3}]$$

$$r_{p1} r_{p1} + r_{p2} r_{p2} + \cancel{r_{p3} r_{p3}} - \cancel{r_{p3} r_{p3}} = r_{p1}^2 \\ x x + y y + z z - z z = x^2 + y^2 = \rho^2$$

$$a \ll b \Rightarrow I_3 \ll I_1$$

$$a \gg b \Rightarrow I_3 \gg I_1$$

More irregular: I_1, I_2, I_3 all different

$$\begin{pmatrix} I_1 & 0 & 0 \\ 0 & I_2 & 0 \\ 0 & 0 & I_3 \end{pmatrix} \Leftrightarrow I_{ij} = I_i \delta_{ij} \quad L_i = \sum_{j=1}^3 I_{ij} \omega_j = \sum_{j=1}^3 I_i \delta_{ij} \omega_j = I_i \omega_i$$

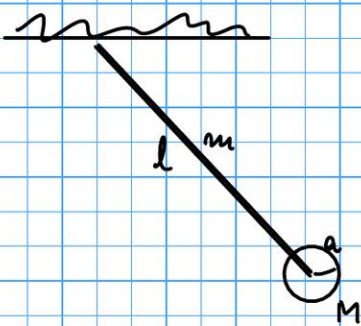
$$\begin{pmatrix} L_1 \\ L_2 \\ L_3 \end{pmatrix} = \begin{pmatrix} I_1 & 0 & 0 \\ 0 & I_2 & 0 \\ 0 & 0 & I_3 \end{pmatrix} \begin{pmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{pmatrix} = \begin{pmatrix} I_1 \omega_1 \\ I_2 \omega_2 \\ I_3 \omega_3 \end{pmatrix}$$

$$T = \frac{1}{2} \vec{\omega} \cdot \vec{L} = \frac{1}{2} \sum_{i=1}^3 \omega_i L_i = \frac{1}{2} \sum_{i=1}^3 \omega_i I_i \omega_i = \frac{1}{2} \sum_{i=1}^3 I_i \omega_i^2$$

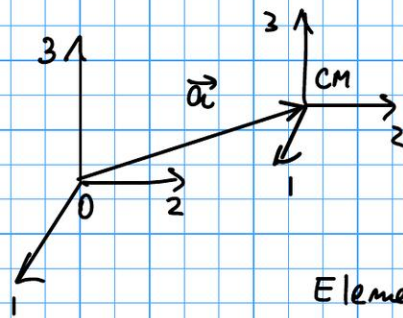
$$L_i = I_i \omega_i$$

$$T = \frac{1}{2} \sum_{i=1}^3 I_i \omega_i^2$$

Parallel Axis THEOREM



($\bar{I}_{ij}$ MoI wrt CM)



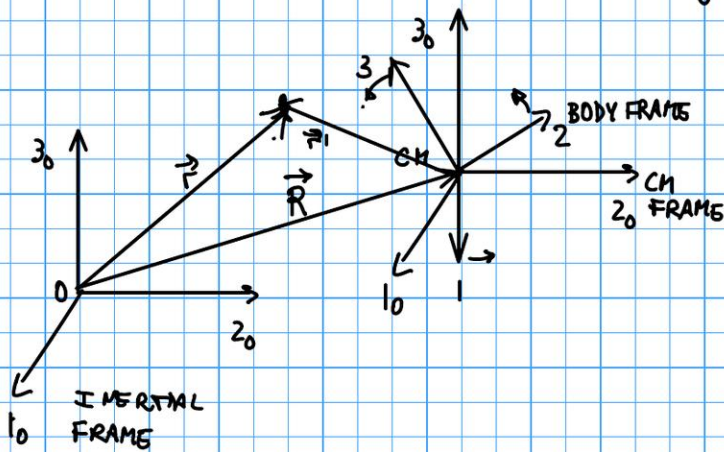
$$I_{ij} = \bar{I}_{ij} + M[s_{ij}a^2 - a_i a_j]$$

$\updownarrow$ ORIGIN O $\updownarrow$ ORIGIN CM

Elementary version: $I = \bar{I} + M d^2$

Rotation wrt an arbitrarily moving point

No longer look at a point in the body fixed in space



- INERTIAL FRAME

- CM FRAME { MAY NOT BE INERTIAL IF $\ddot{\vec{R}} \neq \vec{0}$
AXES // AXES OF INERTIAL FRAME

- BODY FRAME { ORIGIN AT $\vec{R} \Rightarrow$ NON-INERTIAL I $\ddot{\vec{R}} \neq \vec{0}$
AXES $1, 2, 3$ ROTATING WRT $1_0, 2_0, 3_0$

$$\vec{r} = \vec{R} + \vec{r}'$$

$$\left. \frac{d\vec{r}}{dt} \right|_{\text{INERTIAL}} = \frac{d\vec{R}}{dt} + \left. \frac{d\vec{r}'}{dt} \right|_{\text{CM}}$$

$$\left. \frac{d\vec{r}}{dt} \right|_{\text{INERTIAL}} = \frac{d\vec{R}}{dt} + \left. \frac{d\vec{r}'}{dt} \right|_{\text{BODY}} + \vec{\omega} \times \vec{r}'$$

$$\text{IF RIGID BODY} \quad \left. \frac{d\vec{r}}{dt} \right|_{\text{INERTIAL}} = \frac{d\vec{R}}{dt} + \vec{\omega} \times \vec{r}'$$

For any system of particles:

$$M = \sum_p m_p$$

$$\vec{R} \equiv \text{CENTER OF MASS POSITION} \equiv \frac{\sum_p m_p \vec{r}_p}{\sum_p m_p}$$

$$T = \frac{1}{2} M \dot{\vec{R}}^2 + T'$$

$$T' = \sum_p \frac{m_p}{2} \dot{\vec{r}}_p'^2$$

MEASURED IN FRAME
WITH ORIGIN AT CM

$$\vec{L} = \vec{R} \times M \dot{\vec{R}} + \vec{L}'$$

$$\vec{L}' = \sum_p \vec{r}_p' \times m_p \dot{\vec{r}}_p'$$

(CM FRAME)

For rigid body $\dot{\vec{r}}_P' = \vec{\omega} \times \vec{r}_P'$

$$\vec{L} = \vec{R} \times M \dot{\vec{R}} + \underbrace{\sum_P \vec{r}_P' \times m_P (\vec{\omega} \times \vec{r}_P')}_{\text{CALCULATED BEFORE}}$$

$$L_i' \equiv \sum_j \bar{I}_{ij} \omega_j$$

$$T = \frac{1}{2} M |\dot{\vec{R}}|^2 + \underbrace{\sum_P \frac{m_P}{2} (\vec{\omega} \times \vec{r}_P')^2}_{\text{CALCULATED BEFORE}}$$

$$T' \equiv \frac{1}{2} \sum_{ij} \bar{I}_{ij} \omega_i \omega_j$$

EULER'S EQUATIONS

$$\frac{d\vec{L}}{dt} = \vec{N} \quad \text{IF} \left\{ \begin{array}{l} \times \text{ INERTIAL FRAME} \\ \times \text{ CM FRAME (ORIGIN AT CM)} \\ \text{MAY BE ACCELERATED BUT} \\ \text{IT STAYS PARALLEL TO INERTIAL FRAME)} \end{array} \right\} \Leftrightarrow \text{"TRANSLATED BUT NOT ROTATED"}$$

$$\vec{N} = \frac{d\vec{L}}{dt} = \left. \frac{d\vec{L}}{dt} \right|_{\text{BODY}} + \vec{\omega} \times \vec{L}$$

$$N_1 = \left. \frac{dL_1}{dt} \right|_{\text{BODY}} + (\vec{\omega} \times \vec{L})_1 = \left. \frac{dL_1}{dt} \right|_{\text{BODY}} + \omega_2 L_3 - \omega_3 L_2$$

$$L_i = I_i \omega_i \quad \dot{I}_i|_{\text{BODY}} = 0 \quad (\text{RIGID BODY} \Rightarrow \vec{r}_P = \vec{0} \text{ IN BODY FRAME} \Rightarrow \dot{I}_i|_{\text{BODY}} = 0)$$

$$= I_1 \left. \frac{d\omega_1}{dt} \right|_{\text{BODY}} + \omega_2 I_3 \omega_3 - \omega_3 I_2 \omega_2 = I_1 \left. \frac{d\omega_1}{dt} \right|_{\text{BODY}} + \omega_2 \omega_3 (I_3 - I_2)$$

$\frac{d}{dt} \Big|_{\text{BODY}}$

$\frac{d}{dt} \Big|_{\text{BODY}}$

$$I_1 \frac{d\omega_1}{dt} \Big|_{\text{BODY}} = \omega_2 \omega_3 (I_2 - I_3) + N_1$$

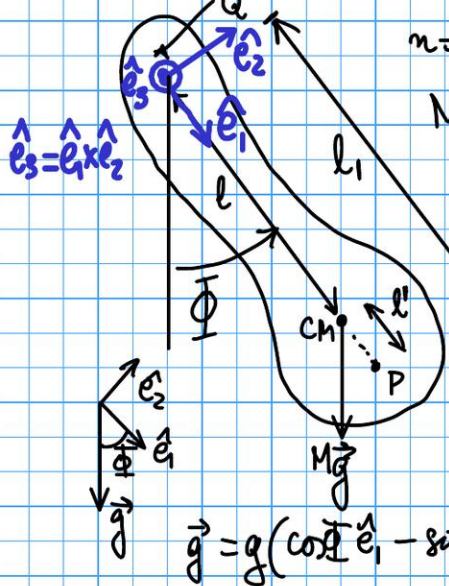
$$I_2 \frac{d\omega_2}{dt} \Big|_{\text{BODY}} = \omega_3 \omega_1 (I_3 - I_1) + N_2$$

$$I_3 \frac{d\omega_3}{dt} \Big|_{\text{BODY}} = \omega_1 \omega_2 (I_1 - I_2) + N_3$$

EULER'S
EQUATIONS

APPLICATIONS

Kater's Pendulum



$n=1$ degree of freedom $\Leftrightarrow$ angle $\Phi \equiv$ gen coord

NO FRICTION

In Lab frame:

$$I \dot{\omega}_3 = \frac{dL_3}{dt} = N_3 = -Mg l \sin \Phi \quad \omega_3 = \dot{\Phi}$$

$$\vec{N}_g = l \hat{e}_1 \times M \vec{g} = l \hat{e}_1 \times Mg (\cos \Phi \hat{e}_1 - \sin \Phi \hat{e}_2) = -Mg l \sin \Phi \hat{e}_3$$

SMALL ANGLE $\sin \Phi \approx \Phi$

$$I \ddot{\Phi} \approx -Mg l \Phi \quad \ddot{\Phi} \approx -\frac{Mg l}{I} \Phi$$

$$\ddot{\Phi} = -\Omega^2 \Phi$$

$$\Omega = \left(\frac{Mg l}{I} \right)^{1/2} \text{ FREQ. OF OSCILLATION}$$

$$\vec{g} = g (\cos \Phi \hat{e}_1 - \sin \Phi \hat{e}_2)$$

Parallel axis theorem

$$I \stackrel{\downarrow}{=} \bar{I} + M l^2 = M \bar{k}^2 + M l^2 = M (\bar{k}^2 + l^2) = M k^2$$

DEFINE RADIUS OF GYRATION k

By $I = M k^2$

DEFINE $l_1 \equiv \frac{k^2}{l} = l + \frac{\bar{k}^2}{l}$

✓ ,

$$D1 \quad I = M R^2$$

$$\Omega^2 = \frac{Mgl}{M(k^2 + l^2)} = \frac{g}{l_1}$$

$$[\Omega^2] = \frac{\cancel{L}/T^2}{\cancel{L}} = \frac{1}{T^2} \Rightarrow [l] = \frac{1}{T^2}$$

COMPARE WITH SIMPLE PENDULUM

$$\Omega = \sqrt{\frac{g}{l}}$$

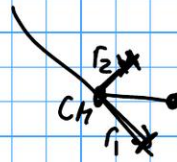


point mass

$$\bar{I} = 0 \Rightarrow k = 0 \quad l_1 = l + \frac{k^2}{l} = l$$

$$\bar{I} = \int d^3r \rho(\vec{r}) (r_1^2 + r_2^2)$$

— r_1^3 $\Rightarrow$ (x^2, y^2)
 distances measured wrt CM



POINT MASS

$$r_1 = r_2 = 0$$

$$\bar{r}^2 = \frac{\bar{I}}{M} = \frac{\int d^3r \rho(\vec{r}) (r_1^2 + r_2^2)}{\int d^3r \rho(\vec{r})} \quad \leftrightarrow \quad \text{Average squared distance to axis of rotation weighted by the mass density } \rho(\vec{r})$$

distance, measured w.r.t. CM

Suspend the pendulum from point P "Percussion center" at distance l_1 from Q, along the direction that goes through CM

$$\Omega_P^2 = \frac{Mgl}{M(\bar{r}^2 + l^2)} = \frac{g}{l_1} \quad \Omega_P^2 = \frac{gl_1}{\bar{r}^2 + l_1^2} = \frac{g(l_1 - l)}{\bar{r}^2 + (l_1 - l)^2} = \frac{g \frac{\bar{r}^2}{l}}{\bar{r}^2 + \bar{r}^4/l^2} = \frac{gl}{l^2 + \bar{r}^2} = \Omega_Q^2$$

$$l_1 = l + \frac{\bar{r}^2}{l} \quad l_1 - l = \frac{\bar{r}^2}{l} \quad (l_1 - l)^2 = \frac{\bar{r}^4}{l^2}$$

Torque - Free Motion in 3D: SYMMETRIC TOP $I_1 = I_2 \neq I_3$ $\Omega \equiv \omega_3 \frac{I_3 - I_1}{I_1}$

$$I_1 \left. \frac{d\omega_1}{dt} \right|_{\text{BODY}} = \omega_2 \omega_3 (I_2 - I_3) \Rightarrow \dot{\omega}_1 = - \left[\omega_3 \frac{(I_3 - I_2)}{I_1} \right] \omega_2 \Rightarrow$$

$$I_2 \left. \frac{d\omega_2}{dt} \right|_{\text{BODY}} = \omega_3 \omega_1 (I_3 - I_1) \Rightarrow \dot{\omega}_2 = \left[\omega_3 \frac{(I_3 - I_1)}{I_2} \right] \omega_1 \Rightarrow$$

$$I_3 \left. \frac{d\omega_3}{dt} \right|_{\text{BODY}} = \omega_1 \omega_2 (I_1 - I_2) \Rightarrow$$

$$\dot{\omega}_1 = -\Omega \omega_2$$

$$\dot{\omega}_2 = \Omega \omega_1$$

$$\dot{\omega}_3 = 0$$

$$\dot{\vec{\omega}} = \Omega \hat{e}_3 \times \vec{\omega} \Leftrightarrow \vec{\omega} \text{ rotates with angular velocity } \Omega \hat{e}_3 \Rightarrow \text{PRECESSION}$$

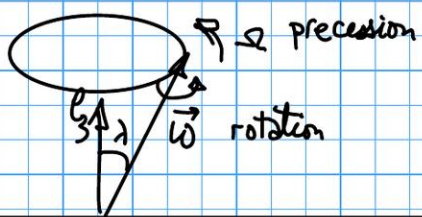
$$\dot{\vec{\omega}} = \Omega \hat{e}_3 \times \vec{\omega} = \Omega \hat{e}_3 \times (\omega_1 \hat{e}_1 + \omega_2 \hat{e}_2 + \omega_3 \hat{e}_3) = \Omega (\omega_1 \hat{e}_2 - \omega_2 \hat{e}_1)$$

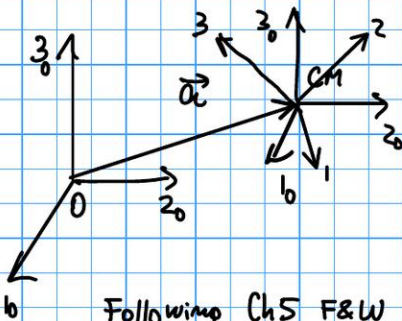
$$\dot{\omega}_1 = -\Omega \omega_2 \quad \dot{\omega}_2 = \Omega \omega_1 \quad \dot{\omega}_3 = 0$$

$\vec{\omega}$ rotates around $\hat{e}_3$ with any speed Ω

If initially $\vec{\omega} = \omega (\sin \lambda \hat{e}_1 + \cos \lambda \hat{e}_3)$ $\vec{\omega} \in$ plane of $\hat{e}_1$ and $\hat{e}_3$

$$\omega_1 = \omega \sin \lambda \cos(\Omega t) \quad \omega_2 = \omega \sin \lambda \sin(\Omega t) \quad \omega_3 = \omega \cos \lambda$$





Following Ch 5 F&W
(Skipping a lot)

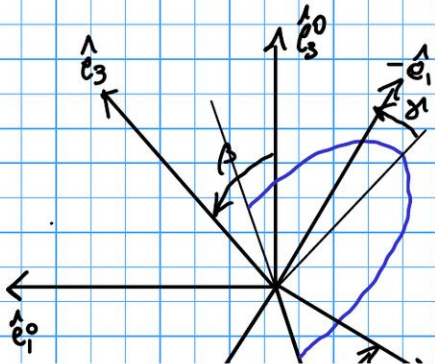
$\hat{e}_1^0, \hat{e}_2^0, \hat{e}_3^0$ INERTIAL SET OF UNIT VECTORS
(NOT ROTATING)

$\hat{e}_1, \hat{e}_2, \hat{e}_3$ BODY SET OF UNIT VECTORS
(ROTATING WITH BODY)

We need 6 coordinates to describe positions of all points
in a RIGID BODY

→ CM position $\vec{R} \leftrightarrow X Y Z$
→ 3 coord to fix orientation of
A 2,2 unit 1,0,0,0,0

Following Ch 5 F&W
(Skipping a lot)

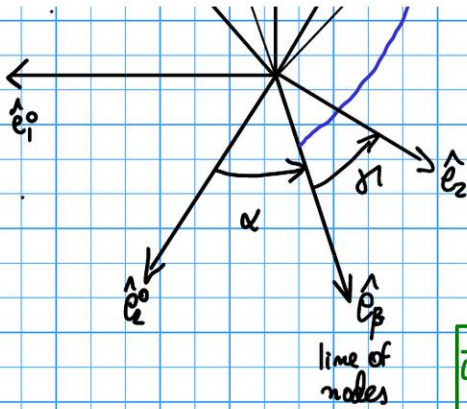


CM position $R \leftrightarrow X Y Z$
3 coord to fix orientation of
 $\hat{e}_1, \hat{e}_2, \hat{e}_3$ wrt $\hat{e}_1^0, \hat{e}_2^0, \hat{e}_3^0$

WE CHOOSE EULER ANGLES: α, β, γ

- $\hat{e}_3 \perp \hat{e}_1, \hat{e}_2$
- ORIENTATION OF NEW NORTH POLE $\rightarrow \beta$ POLAR $\rightarrow \alpha$ AZIMUTHAL
- γ SPECIFIES ADDITIONAL ROTATION AROUND $\hat{e}_3$
- α, β, γ CAN BE CHOSEN INDEPENDENTLY AND THEY COMPLETELY SPECIFY ORIENTATION OF BODY FRAME $\Rightarrow$ USE α, β, γ AS GENERALIZED COORD

T-16.2.1 + .2.1 + .2.1 (1)



• 3-DOF ROTATION ADDITIONAL NOTATION INVOLVED

• α, β, γ CAN BE CHOSEN INDEPENDENTLY AND THEY COMPLETELY SPECIFY ORIENTATION OF BODY FRAME $\Rightarrow$ USE α, β, γ AS GENERALIZED COORD

$$T = \frac{1}{2} (\mathcal{I}_1 \omega_1^2 + \mathcal{I}_2 \omega_2^2 + \mathcal{I}_3 \omega_3^2) \quad (1)$$

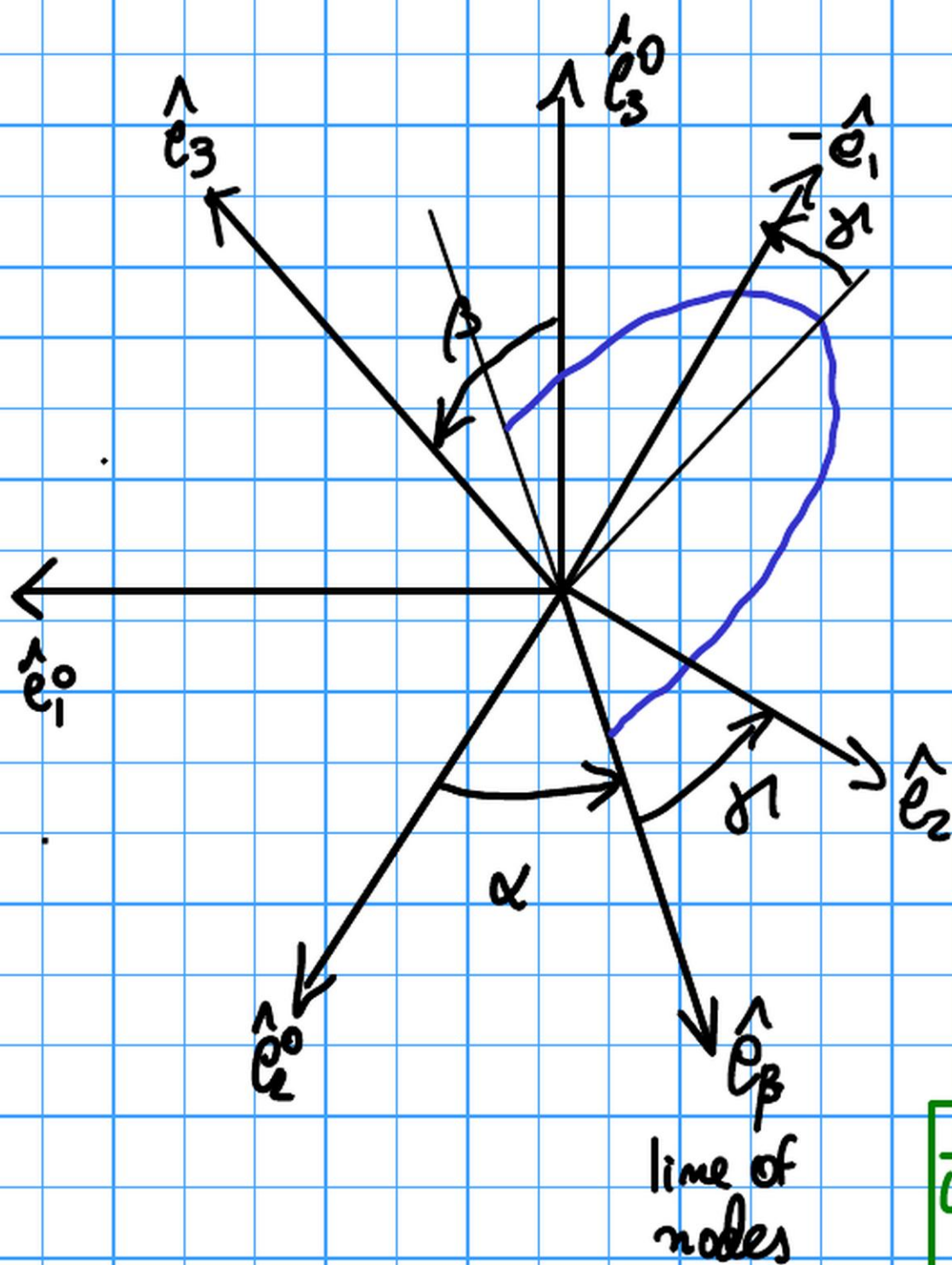
$$\vec{\omega} = \alpha \hat{e}_3^0 + \beta \hat{e}_3^B + \dot{\gamma} \hat{e}_3 \quad (2) \quad \hat{e}_3^B = \cos \gamma \hat{e}_2 + \sin \gamma \hat{e}_1 \quad (3)$$

$$\hat{e}_3^0 = \cos \beta \hat{e}_3 + \sin \beta [\cos \gamma (-\hat{e}_1) + \sin \gamma \hat{e}_2] \quad (4)$$

$$\vec{\omega} = (-\dot{\alpha} \sin \beta \cos \gamma + \dot{\beta} \sin \gamma) \hat{e}_1 + (\dot{\alpha} \sin \beta \sin \gamma + \dot{\beta} \cos \gamma) \hat{e}_2 + (\dot{\gamma} + \dot{\alpha} \cos \beta) \hat{e}_3 \quad (5)$$

b

Following Ch5 F&W
(Skipping a lot)



NOTE:

Infinitesimal rot

$$R_k = 1 + M_k \dot{\alpha} dt \quad (6)$$

$$R_\alpha R_\beta = 1 + (M_\alpha \dot{\alpha} + M_\beta \dot{\beta}) dt + \cancel{O(dt^2)} \quad (8)$$

$$R_\beta = 1 + M_\beta \dot{\beta} dt \quad (7)$$

$$R_\beta R_\alpha = 1 + (M_\beta \dot{\beta} + M_\alpha \dot{\alpha}) dt + \cancel{O(dt^2)} \quad (9)$$

EQUAL TO ORDER dt

$$\Rightarrow \text{INFITESIMAL ROTATIONS ADD, } \vec{\omega} = \vec{\omega}_x + \vec{\omega}_y + \vec{\omega}_z \quad (10) \Rightarrow (2)$$

KINETIC ENERGY

General case:

KINETIC ENERGY

General case:

$$T = \frac{I_1}{2} \omega_1^2 + \frac{I_2}{2} \omega_2^2 + \frac{I_3}{2} \omega_3^2 = \frac{I_1}{2} (-\dot{\alpha} \sin \beta \cos \gamma + \dot{\beta} \sin \gamma)^2 + \frac{I_2}{2} (\dot{\alpha} \sin \beta \sin \gamma + \dot{\beta} \cos \gamma)^2 + \frac{I_3}{2} (\dot{\gamma} + \dot{\alpha} \cos \beta)^2 \quad (11)$$

Symmetric Top:

$$I_1 = I_2 \quad (12)$$

$$T = \frac{I_1}{2} (\dot{\beta}^2 + \dot{\alpha}^2 \sin^2 \beta) + \frac{I_3}{2} (\dot{\gamma} + \dot{\alpha} \cos \beta)^2 \quad (13)$$

Torque-Free Motion of symmetric top

$$L = T = \frac{I_1}{2} (\dot{\beta}^2 + \dot{\alpha}^2 \sin^2 \beta) + \frac{I_3}{2} (\dot{\gamma} + \dot{\alpha} \cos \beta)^2 \quad (14)$$

α, β, γ Gen Coord $\Rightarrow$ 3 Lagrange Eqs

2 CYCLIC COORDINATES: α, γ (15)

$$\Rightarrow \begin{cases} 2 \text{ CONST OF MOTION} \\ p_\alpha = \frac{\partial L}{\partial \dot{\alpha}} = I_1 \sin^2 \beta \dot{\alpha} + I_3 (\dot{\gamma} + \dot{\alpha} \cos \beta) \cos \beta \quad (16) \\ p_\gamma = \frac{\partial L}{\partial \dot{\gamma}} = I_3 (\dot{\gamma} + \dot{\alpha} \cos \beta) \quad (17) \end{cases}$$

WE WILL SHOW LATER THAT:

$$p_\alpha = \vec{L} \cdot \hat{e}_3^0 \quad (18)$$

$$p_\gamma = \vec{L} \cdot \hat{e}_3 \quad (19)$$

recall $\vec{\omega} = (-\dot{\alpha} \sin \beta \cos \gamma + \dot{\beta} \sin \gamma) \hat{e}_1 + (\dot{\alpha} \sin \beta \sin \gamma + \dot{\beta} \cos \gamma) \hat{e}_2 + (\dot{\gamma} + \dot{\alpha} \cos \beta) \hat{e}_3$ (5)

We compute now $p_{\gamma} = \frac{\partial L}{\partial \dot{\gamma}} = I_3 (\dot{\gamma} + \dot{\alpha} \cos \beta)$ (17) and notice that $p_{\gamma} \stackrel{(5)}{=} I_3 \omega_3 = \vec{L} \cdot \hat{e}_3$ (19) ✓
 AS ANTICIPATED

Also $p_{\alpha} = \frac{\partial L}{\partial \dot{\alpha}} = I_1 \sin^2 \beta \dot{\alpha} + I_3 (\dot{\gamma} + \dot{\alpha} \cos \beta) \cos \beta$ (16) $\vec{L} = I_1 (\omega_1 \hat{e}_1 + \omega_2 \hat{e}_2) + I_3 \omega_3 \hat{e}_3$ (20)

Compare with $\vec{L} \cdot \hat{e}_3 \stackrel{(20)}{=} (I_1 (\omega_1 \hat{e}_1 + \omega_2 \hat{e}_2) + I_3 \omega_3 \hat{e}_3) \cdot (\cos \beta \hat{e}_3 + \sin \beta [\cos \gamma (-\hat{e}_1) + \sin \gamma \hat{e}_2])$ (21)

$$= I_1 [(-\dot{\alpha} \sin \beta \cos \gamma + \dot{\beta} \sin \gamma)(-\sin \beta \cos \gamma) + (\dot{\alpha} \sin \beta \sin \gamma + \dot{\beta} \cos \gamma)(\sin \beta \sin \gamma)]$$

$$+ I_3 (\dot{\gamma} + \dot{\alpha} \cos \beta) \cos \beta$$
 (22)

$$= I_1 \dot{\alpha} \sin^2 \beta (\omega^2 \gamma + \sin^2 \gamma) + I_3 (\dot{\gamma} + \dot{\alpha} \cos \beta) \cos \beta \quad (23)$$

$$\boxed{\vec{L} \cdot \hat{e}_3 = I_1 \dot{\alpha} \sin^2 \beta + I_3 (\dot{\gamma} + \dot{\alpha} \cos \beta) \cos \beta = p_\alpha} \quad \checkmark \quad (18)$$

$$p_\beta = \frac{\partial L}{\partial \dot{\beta}} = I_1 \dot{\beta} \quad (24) \text{ NOT A CONSTANT (IN PRINCIPLE) BECAUSE } \beta \text{ APPEARS EXPLICITLY IN THE LAGRANGIAN}$$

$$\frac{\partial L}{\partial \beta} = I_1 \sin \beta \cos \beta \dot{\alpha}^2 - I_3 (\dot{\gamma} + \dot{\alpha} \cos \beta) \dot{\alpha} \sin \beta = \dot{\alpha} \sin \beta (I_1 \dot{\alpha} \cos \beta - I_3 \omega_3) \quad (25)$$

Then

$$\dot{p}_\beta = \frac{\partial L}{\partial \beta} = \dot{\alpha} \sin \beta (I_1 \dot{\alpha} \cos \beta - I_3 \omega_3) \quad (26) \quad (\text{However, see later})$$

Compare p_β with $\vec{L} \cdot \hat{e}_\beta = \overset{(20)}{\rightarrow} (I_1 (\omega_1 \hat{e}_1 + \omega_2 \hat{e}_2) + I_3 \omega_3 \hat{e}_3) \overset{(3)}{\rightarrow} (\cos \gamma \hat{e}_2 + \sin \gamma \hat{e}_1)$

Recall $\vec{\omega} = (-\dot{\alpha} \sin \beta \cos \gamma + \dot{\beta} \sin \gamma) \hat{e}_1 + (\dot{\alpha} \sin \beta \sin \gamma + \dot{\beta} \cos \gamma) \hat{e}_2 + (\dot{\gamma} + \dot{\alpha} \cos \beta) \hat{e}_3 \quad (5)$

$$\begin{aligned}\vec{L} \cdot \hat{e}_\beta &= I_1 \left[(-\dot{\alpha} \sin \beta \cos \gamma + \dot{\beta} \sin \gamma) \sin \gamma + (\dot{\alpha} \sin \beta \sin \gamma + \dot{\beta} \cos \gamma) \cos \gamma \right] \quad (27) \\ &= I_1 \dot{\beta} (\sin^2 \gamma + \cos^2 \gamma) = I_1 \dot{\beta} \Rightarrow \boxed{\vec{L} \cdot \hat{e}_\beta = p_\beta} \quad (29) \quad \checkmark\end{aligned}$$

Description in Inertial (or CM) Frame

$$\begin{aligned}\text{No torque} \Rightarrow \dot{\vec{L}} &= \vec{0} \quad (30) \\ \Rightarrow \vec{L} &= \vec{0} \quad (31) \\ \Rightarrow \text{CHOOSE } \hat{e}_3^0 // \vec{L} \quad (32) &\Rightarrow \vec{L} \cdot \hat{e}_3^0 = |\vec{L}| \quad (34) \text{ AND } \vec{L} = |\vec{L}| \hat{e}_3^0 \quad (35) \\ \Rightarrow |\vec{L}| \text{ doesn't change} \quad (33)\end{aligned}$$

We know that $\vec{L} \cdot \hat{e}_3 = p_\phi$
is a constant of the motion

$$\underbrace{\vec{L} \cdot \hat{e}_3}_{= p_\phi = \text{CONST}} = \underbrace{|\vec{L}|}_{= \text{CONST}} \underbrace{\hat{e}_3^0 \cdot \hat{e}_3}_{= \cos \alpha \quad (36)} \Rightarrow \hat{e}_3^0 \cdot \hat{e}_3 = \text{CONSTANT} \quad (37)$$

= CONST

$$\text{CONST} = \hat{e}_3^0 \cdot \hat{e}_3^1 = \cos(\text{angle between } \hat{e}_3 \text{ and } \hat{e}_3^0) = \cos \beta \quad (38)$$

β IS A CONSTANT OF THE MOTION $\Rightarrow \dot{\beta} = 0 \Rightarrow$ (39) (40)

$$p_\beta = I, \dot{\beta} = \vec{L} \cdot \hat{e}_\beta = 0 \quad (41)$$

For this particular choice of axes, p_β and $\vec{L} \cdot \hat{e}_\beta$ are zero!!

$$\begin{aligned} (41) \quad \vec{L} \cdot \hat{e}_\beta &= 0 \\ \vec{L} &\neq 0 \quad |\hat{e}_\beta| = 1 \neq 0 \end{aligned}$$

$$\Rightarrow \vec{L} \perp \hat{e}_\beta \Rightarrow \quad (42)$$

$$\vec{L} \text{ is in the plane of } \hat{e}_3 \text{ and } \hat{e}_3^0 \quad (43)$$

CONSTANTS: $I_1, I_3, \beta, I_3 \omega_3 = L_3, p_\alpha$

$$p_\alpha = \frac{\partial L}{\partial \dot{\alpha}} = I_1 \sin^2 \beta \dot{\alpha} + I_3 (\dot{\gamma} + \dot{\alpha} \cos \beta) \cos \beta \quad (16) \quad (5) \Rightarrow \underline{I_1 \sin^2 \beta \dot{\alpha} + I_3 \omega_3 \cos \beta} \quad (44) \Rightarrow \boxed{\dot{\alpha} \text{ CONSTANT}} \quad (45)$$

$$p_\gamma = \frac{\partial L}{\partial \dot{\gamma}} = I_3 (\dot{\gamma} + \dot{\alpha} \cos \beta) \quad (17) \Rightarrow \boxed{\dot{\gamma} \text{ CONSTANT}} \quad (46)$$

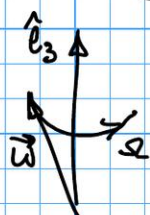
$I_3, p_\alpha, \dot{\alpha}, \beta \text{ CONSTANT}$

$\alpha = \dot{\alpha}_0 t$
 $(47) \uparrow$

$\gamma = \dot{\gamma}_0 t$
 $\nearrow \text{CONSTANTS} \quad (48)$

$$(41) \downarrow 0 = \dot{p}_\beta = \frac{\partial L}{\partial \beta} \stackrel{(26)}{=} \dot{\alpha} \sin \beta (I_1 \dot{\alpha} \cos \beta - I_3 \omega_3) \Rightarrow \boxed{\dot{\alpha} \cos \beta = \frac{I_3}{I_1} \omega_3} \quad (49)$$

$$(5) \Rightarrow \omega_3 = \dot{\gamma} + \dot{\alpha} \cos \beta \stackrel{(49)}{=} \dot{\gamma} + \frac{I_3}{I_1} \omega_3 \Rightarrow \boxed{\dot{\gamma} = \omega_3 \frac{I_1 - I_3}{I_1} = -\Omega} \quad (51)$$



$$\Omega = \omega_3 \frac{I_3 - I_1}{I_1} \quad (50)$$

defined when looking at body frame evolution of $\vec{\omega}$ a couple of lectures before

$$\stackrel{(2)}{(40)} \Rightarrow \boxed{\vec{\omega} = \dot{\alpha} \hat{e}_3^0 + \dot{\gamma} \hat{e}_3} \stackrel{(52)}{\Rightarrow} \boxed{\hat{e}_3, \hat{e}_3^0, \vec{\omega}, \vec{e}} \text{ are in one plane} \quad (53)$$

Consider cases where $I_1 = I_2 < I_3$ or $I_1 = I_2 > I_3$

$$\vec{L} = I_1 (\omega_1 \hat{e}_1 + \omega_2 \hat{e}_2) + I_3 \omega_3 \hat{e}_3 \quad (20)$$

SIMPLE PENDULUM:

$$\ddot{\theta} = -\frac{g}{l} \sin \theta$$

THUMB TACK:

$$(\text{angle})'' = - \left[\begin{array}{c} \text{AWFUL} \\ \text{MESS} \end{array} \right] \sin(\text{angle})$$
